PRODUCT CIRCUIT DIAGRAM

A SYSTEMS VIEW OF HOW VALUE FLOWS THROUGH A PRODUCT — REVEALING FRICTION, MISMATCH, AND OPPORTUNITY

> *"Utilities are sort of impedance-matched with the government. If you look at a utility, they are heavily regulated by public utility commissions.So they're sort of impedance-matched with the government—both literally and figuratively"* — ***Elon Musk***

🧠 CORE ANALOGY

Electrical Concept	Symbol	Product/System Equivalent
Voltage (V)	⚡⚡⚡	Strength of User Intent / Business Demand
Current (I)	—	Flow of Value / progress
Resistance (R)	-WW-	Friction (UX, process, org)
Impendance / Mismatch		System Misalignment (handoffs,formats etc.)
Capacitor / Delay	⊣⊢	Waiting / Buffering (queues, delays)
Amplifier	▷	Simplify / Automate / AI
Blackbox	■	Stable, trusted subsystem
Whitebox	□	Problem area to inspect
Measurement		Outcome (time, cost, success rate)

⚡ CREATING VALUE ⚡

Outcome Quality ∝ (Flow − Resistance − Mismatch − Delay) + Amplification

THE UNLOCK 🔓

Instead of asking:

"What feature should we build?"

Ask:

"Where is energy being lost in the system?"

SAP Institute for Product and Engineering

Rana Chakrabarti | 2026

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EXAMPLE: SMALL BUSINESS LOAN APPLICATION

Intent: Get loan approved quickly

→ Start Application
→ → Long form, unclear fields ▷ AI document parser

→ Upload Documents
→ → File format confusion ▷ Auto-validation of inputs

→ Mismatch:
→ → User uploads PDFs → System expects structured data

→ Verification
→ → 24-48 hr delay ▷ Real-time status tracking

→ Manual Review
→ → Duplicate work

→ Decision

→ Outcome
→→ 5 days
→→ High cost
→→ 60% completion rate

CIRCUIT COMPONENTS

Flow (wire)
Value / progress moving through the system

White Box- Inspect
- Problem area
- Break into sub-steps

Black Box- Trust
- Working fine
- Do not expand

Resistance / Friction
- UX friction
- Process friction
- Org friction

Impedance / Mismatch
- System handoffs
- Data mismatches
- Incentives/expectation mismatches

Capacitor / Delay
- Waiting
- Queues
- Batch Processing

Amplifier
- Simplification
- Automation
- AI Agents

Measurements / Outcomes
- ⏱ Time
- $ Cost
- ✓ Success rate

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HOW TO BUILD (STEP-BY-STEP)

Facilitation Questions

1. DEFINE INTENT (VOLTAGE) ⚡

What is the user trying to achieve?

- What job is the user renting the process for?
- What is the intensity of the intent?

2. DRAW THE FLOW

5–9 steps from intent → outcome

- What did the end-user describe as their experience?
- What’s happening beneath the hood?

3. ADD RESISTANCE

Where is it hard, slow, or frustrating?

- Where do users struggle?
- Where do teams to extra work?

4. MARK MISMATCHES

Where do things not align?

- Where do systems not align ?
- Where do expectations break?

5. ADD DELAYS

Where does time accumulate?

- Where are there queues, buffers or delays?
- What waits on what?

6. OPEN ONE WHITE BOX

Zoom into the biggest problem area

- Which processes/steps have the most opportunity?
- Consider existing user research where needed

7. ADD AMPLIFIERS

Where can we reduce loss or increase flow?

- What can be simplified?
- What can be automated?
- What can be made intelligent?

8. MEASURE OUTCOMES

What improves?

- What is being measured?
- What marks success?

Outcome Quality ∝ (Flow − Resistance − Mismatch − Delay) + Amplification

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